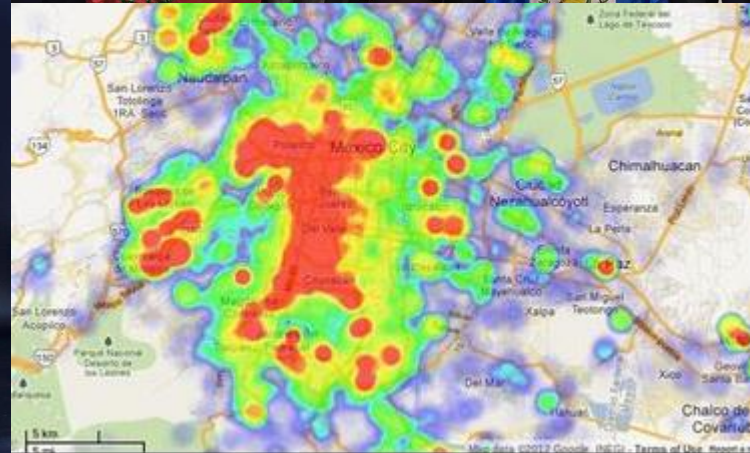
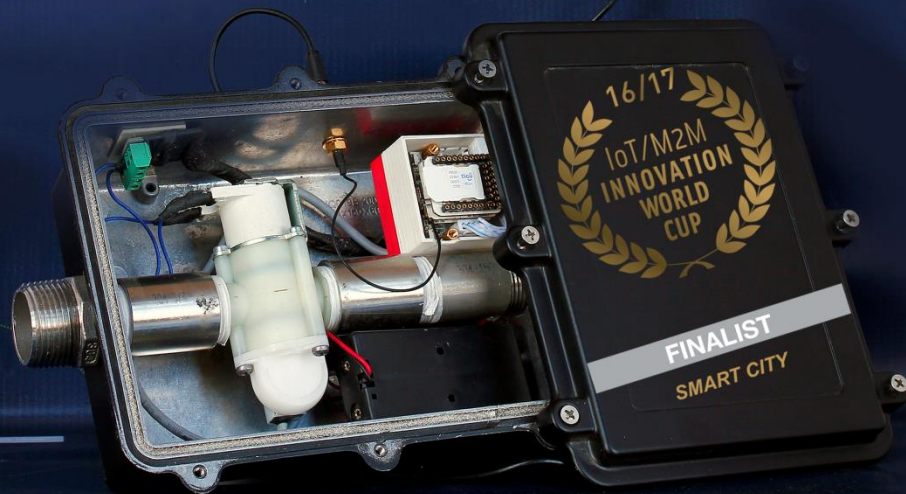


# Enabling, Autonomous Urban Water Utilities



WaterBox,  
IoT Context-Aware and Self Sufficient

# Flatbox



SOLVE



SOLVE

# Why? Autonomous Urban Water Utilities

Automatic  
user  
profiling



Automatic  
User  
Mapping



Real time  
Service  
Monitoring



Real Time  
Resource  
Management



Extend Service  
Pre-paid,  
PAYG, etc.



Lowers  
Cost of  
Operations



Lowers  
Cost of  
Distribution



Enhances  
Fault  
Detection



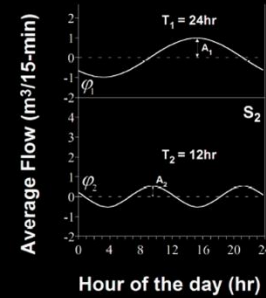
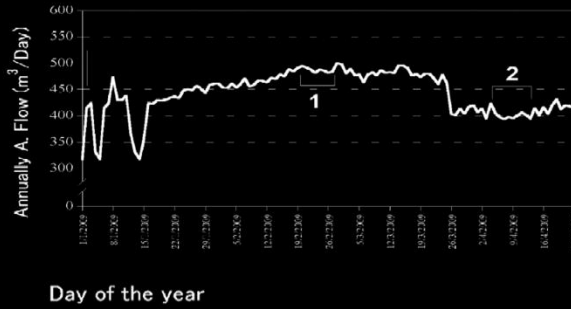
Enhances  
Conservation  
Efforts



Enhances  
Transparency

# How? Autonomous Urban Water Utilities

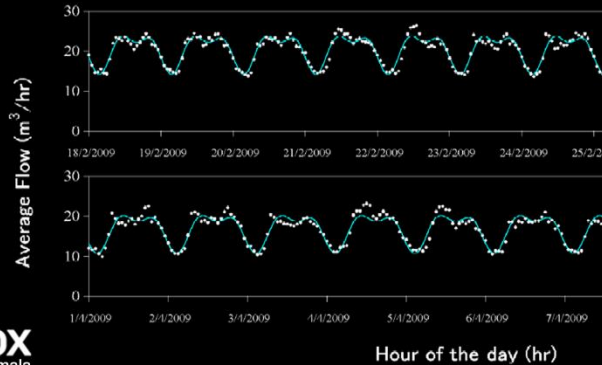
## 1. IoT Sends Real Time Data



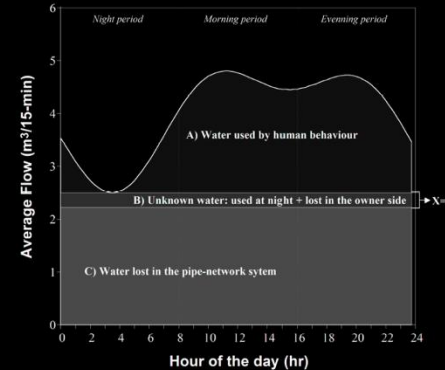
**3. Utility Consumption Signature is generated, compared, and mapped in real time**

$$W(t) = \bar{w} + A_1 \sin\left(\frac{2\pi}{T_1}t + \varphi_1\right) + A_2 \sin\left(\frac{2\pi}{T_2}t + \varphi_2\right)$$

## 2. Model is applied to data stream



**Flatbox**  
campus TEC, Guatemala



**4. Big data is analyzed to generate leads on resource management**

**SOLVE**

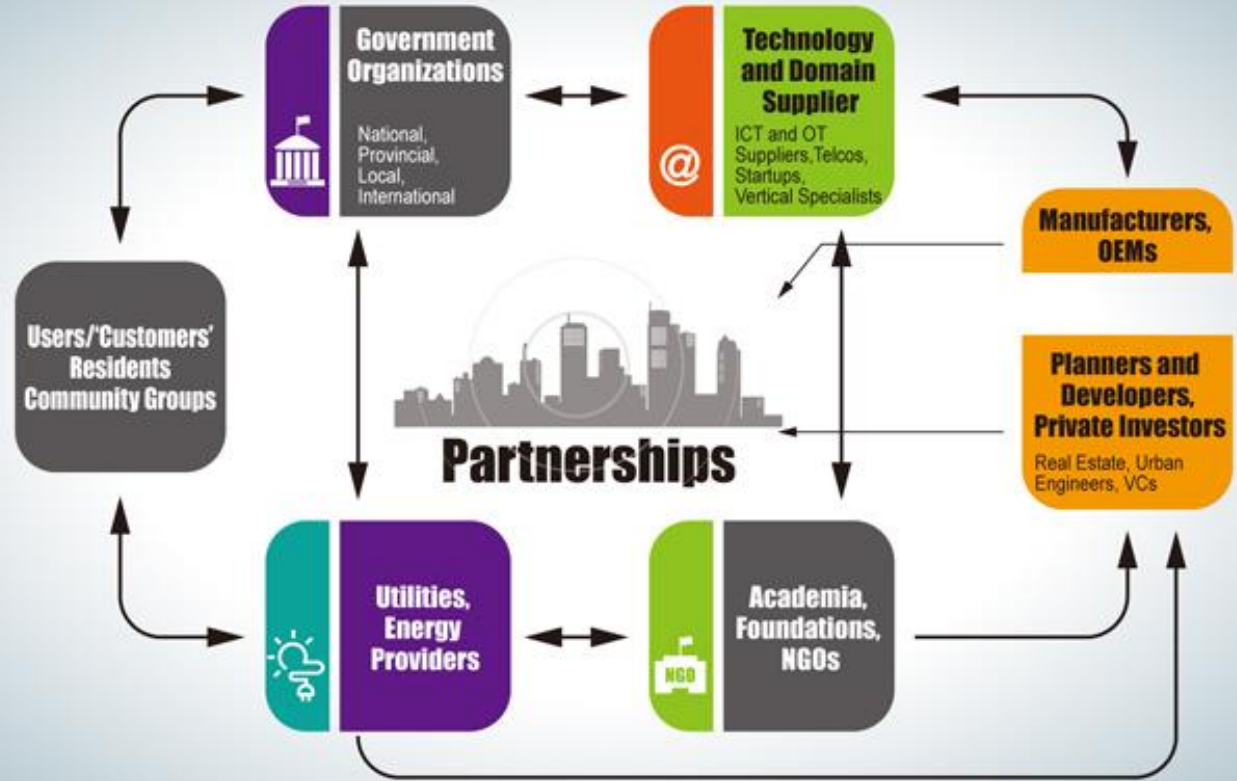
Idea



Device



Solution



SOLVED!